firepower of cosmic rays, is the notion that a powerful enough collision could trigger a universe-destroying quantum event called *vacuum decay*. The whole idea of vacuum decay rests on the hypothesis that our universe has a kind of fatal instability built into it. While that might sound scary even as just as a remote possibility, at the time the RHIC was commissioned, there was no real evidence for such a flaw, so it wasn't taken especially seriously.

When the LHC discovered the Higgs boson, in 2012, that all changed.

THE STATE OF THE UNIVERSE

A good way to make a particle physicist cringe is to refer to the Higgs boson by the name that made it famous: the *God Particle*. Our collective grumpiness around this lofty moniker isn't fueled entirely by a discomfort with the mixing of science and religion (though for many that's a big part of it). It's also that "God Particle" is just terribly imprecise, and sounds a bit presumptuous, frankly. Which is not to say that the Higgs boson isn't a deeply important part of the Standard Model of particle physics. It could even be argued that the Higgs is key to everything else fitting together. But it's really the Higgs *field*, not the particle, that plays a central role in the workings of particle physics and the nature of the cosmos.

The short version of the story is that the Higgs is a kind of energy field that pervades all of space and has interactions with other particles in a way that allows them to have mass. The Higgs *boson* has the same relationship to the Higgs field that the photon, the carrier of the electromagnetic force (and light), has to the electromagnetic field—it's a localized "excitation" of something that pervades a larger space. The long version of the story has to do with electroweak theory, the theory that unites the weak nuclear force with electricity and magnetism, and how a process called "spontaneous symmetry breaking" separates those forces.

(This is the part of the book where I really really want to teach you all of quantum field theory, but where by a heroic effort I limit myself to just touching on a few key issues. You'll just have to trust me that if you decide to go and learn the mathematics behind all this, it gets MUCH cooler.)

We talked in Chapter 2 about the fact that physics works differently at different energies. Electromagnetism and the weak nuclear force, for instance, act like fully separate phenomena at the kinds of energies we deal